

Exam - I
CSCE 584 : Neural Networks and Their Applications
Fall 2025
Due - October 19, End of Day

Instructions:

- Do not use any optimization package for implementing the optimization algorithm/training for any problem.
- If the answers are not readable, they will not be graded
- Points will be deducted if a problem is not solved as well as presented (solutions) in its entirety. Pay attention to details.
- If you refer to any resource to get your solutions, add an acknowledgement with your solutions (details of the source, e.g., book, website, Git repository, etc.).
- Include codes to the coding problems (share github or Colab link) but add captioned figures in the PDF. If the figures are not added as part of the solution, the problem will not be graded.
- Do not include colab link or .ipynb file in your submission without extracting and summarizing the results in your solution document.
- Compile all your solutions into a single file (.pdf) and upload it to Blackboard.
- Name your submission file using this format "Lastname-CSCE584-Exam-1.pdf".

Part A: (Coding) [Choose any three problems]

1. From the research article (a), reproduce the results of Examples 1 and 3. Include a brief summary of the example problems, its formulation, the type of NN architecture considered, and the training process.
 2. From the research article (a), reproduce the results of Example 7. Include a brief summary of the problem, its formulation, the type of NN architecture considered, and the training process.
 3. Write down the forward pass and backward pass equations for a three layer convolution neural network (CNN). You are free to pick the design parameters (i.e., input/output dimension, number of kernels, their size, dilation, etc.) of your choice. You should provide the list of all design parameters for a CNN in general and the design parameters for your network. Define all the parameters/notations and their dimensions.
 4. For the MNIST dataset used in your HW 1 (see link to article below), write down a custom code to train a CNN with three convolutional layers for the handwritten digit classification problem. Article: [Classification of handwritten digits](#) – this is the same code you have submitted for a HW 1 problem.
 5. Use the code for digit classification in HW1 from the article (or your solution to Problem 3 of this exam): [Code for classification of handwritten digits](#). Instead of using the MNIST dataset, use the CIFAR-10 dataset (to perform 10 class classification). Download/Use the dataset from the link given here: CIFAR dataset.
- (a) Kumpati, S.N. and Kannan, P., 1990. Identification and control of dynamical systems using neural networks. IEEE Transactions on neural networks, 1(1), pp.4-27. ([Download PDF here](#))